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URBAN DRAINAGE SYSTEM

A Project for Sustainable Urban Spaces

AI Hackathon 2026



INTRODUCTION

- Nowadays, blocked drainage systems everywhere easily cause severe water overflowing and urban flooding. When heavy rain comes, trash blocks the drains and water cannot flow away quickly.
- Urban drainage system is a simple, smart system that helps clean drains and stops flooding before it happens.



PROJECT BACKGROUND



- Urban flooding happens often because traditional drain systems cannot monitor water levels or trash blockages in real time.
- When heavy rain comes, blocked drains cause water overflow and damage homes and roads.
- Our project introduces a Smart Urban Drainage System where users can easily report blocked or overflowing drains by uploading a photo through our website . In return , users earn reward points for their efforts to keep the city clean.

OBJECTIVES

- Easy Photo Reporting: Let citizens report blocked drains easily using the camera.
- Save Time and Costs : Help maintenance teams fix real drain problems quickly and efficiently.



PROJECT IMPLEMENTATION

- For the implementation , we focused on creating a system that is both easy to use and useful in real-world situations.
- Instead of information being delayed or lost, the problem can be reported, identified , and communicated through one connected system.

PROJECT IMPACT

- Flood Prevention and Safety: Reduces city flooding through real-time sensor alerts and quick blockage cleaning before water overflows.
- Environmental Protection: Prevents plastic trash and waste from flowing into natural rivers and oceans.
- Cost and Time Efficiency: Helps municipal teams focus only on blocked drains, saving labor, fuel, and maintenance costs.
- Citizen Participation: Encourages the public to active community care using the in-app camera to report drain issues.

CONCLUSION

- Urban drainage system provides a smart, modern solution to urban flooding by camera reporting.
- By enabling active participation from citizens , our system protects local infrastructure, saves maintenance costs, and builds a cleaner , safer, and more resilient smart city.

THANK YOU

